

KOCHUBEY, Russian FederationWMO: **370850**Lat: **44.400N**Lon: **46.550E**Elev: **-70**StdP: **14.73**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.0	10.4	-1.1	5.2	3.6	6.1	7.5	11.7	23.9	37.4	21.0	35.9	4.9	300	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	95.6	70.7	92.9	70.2	90.5	69.7	75.2	87.4	73.7	85.9	72.5	84.4	10.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.8	117.2	80.4	70.1	110.6	79.4	68.6	104.7	78.4	38.6	87.9	37.3	85.3	36.1	84.6	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.1	20.0	17.1	-2.3	100.7	9.4	3.4	-9.1	103.1	-14.6	105.1	-19.9	107.0	-26.8	109.4	DB
			-2.8	78.9	9.4	1.2	-9.5	79.8	-15.0	80.5	-20.3	81.2	-27.1	82.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	31.1	32.1	41.1	52.5	65.2	74.7	79.1	78.0	68.0	55.5	43.0	35.3	
	DBStd	18.78	9.42	9.79	6.52	6.82	6.58	5.04	4.43	5.30	6.32	8.18	8.45	8.58	
	HDD50	2150	587	502	284	47	2	0	0	0	40	232	456		
	HDD65	5097	1052	922	741	377	78	2	0	40	306	659	920		
	CDD50	3888	0	0	8	123	473	740	901	869	541	209	23	1	
	CDD65	1363	0	0	0	3	85	293	436	404	131	11	0	0	
	CDH74	13783	0	0	1	49	727	2836	4863	4372	896	39	0	0	
	CDH80	5908	0	0	0	6	182	1122	2335	2036	224	3	0	0	
Wind	WSAvg	7.1	6.8	7.3	7.9	7.4	7.1	7.1	6.9	6.8	7.1	6.7	6.9	7.2	
Precipitation	PrecAvg	11.1	0.7	0.6	0.7	0.9	1.3	1.5	1.1	0.9	1.1	0.9	0.8	0.8	
	PrecMax	15.7	1.3	1.7	1.8	2.2	4.1	4.7	3.4	3.0	3.6	2.6	2.0	1.7	
	PrecMin	6.7	0.1	0.2	0.0	0.0	0.1	0.2	0.2	0.1	0.0	0.0	0.0	0.2	
	PrecStd	2.3	0.3	0.4	0.5	0.6	0.7	1.1	0.9	0.8	1.0	0.7	0.6	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.5	53.4	67.0	80.1	89.0	95.6	100.0	99.0	91.6	78.8	64.1	54.8	
		MCWB	44.7	46.1	52.1	59.6	65.5	69.8	71.9	71.4	72.0	65.3	56.2	50.3	
	2%	DB	46.0	48.5	58.9	74.1	84.9	92.3	95.8	95.6	85.3	73.6	59.6	50.6	
		MCWB	42.6	43.2	49.4	58.2	64.6	69.2	70.8	70.8	68.4	63.3	54.5	47.6	
	5%	DB	43.6	45.1	54.1	69.4	81.2	89.2	93.1	92.8	81.8	70.1	56.8	47.5	
		MCWB	40.9	41.6	46.7	56.3	63.6	67.5	69.8	71.0	67.1	61.4	52.9	44.8	
	10%	DB	41.4	42.6	50.5	65.0	77.9	86.4	90.7	90.0	78.9	66.9	54.3	45.1	
		MCWB	39.6	40.2	44.6	54.4	62.4	66.6	69.7	70.7	65.6	60.1	51.0	43.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	47.5	53.7	61.6	69.2	73.9	76.9	77.3	74.8	66.7	58.1	51.3
			MCDB	49.0	52.9	64.9	74.3	81.2	88.5	91.7	88.7	86.0	74.7	61.7	53.6
2%		WB	43.0	44.0	50.0	59.4	67.1	72.0	74.8	75.7	71.8	64.8	55.4	47.8	
		MCDB	45.5	47.3	57.7	72.2	79.3	85.9	88.3	87.0	81.0	71.8	58.9	50.4	
5%		WB	41.3	42.1	47.4	57.1	65.4	70.6	73.3	74.3	69.3	62.9	53.4	45.0	
		MCDB	43.1	44.5	53.0	67.4	76.9	83.3	86.0	85.7	78.1	68.5	56.5	46.8	
10%		WB	39.6	40.5	45.3	55.2	63.8	69.1	72.0	72.9	67.6	60.8	51.3	43.2	
		MCDB	41.2	42.4	49.6	63.6	75.1	81.2	84.3	84.4	76.1	65.9	53.9	44.9	
Mean Daily Temperature Range		5% DB	MDBR	9.7	11.1	13.5	17.6	19.1	19.7	19.8	20.3	17.8	15.1	11.6	9.4
			MCDBR	12.9	15.0	20.1	25.2	23.6	23.5	23.7	24.0	20.5	17.9	15.2	12.0
	MCWBR		9.4	10.1	11.7	12.1	7.9	7.4	7.0	7.6	8.3	9.1	10.3	9.1	
	5% WB	MCDBR	11.5	12.8	18.6	22.3	21.3	20.8	21.0	21.4	18.0	16.8	13.8	10.6	
		MCWBR	8.9	9.4	11.4	11.5	8.5	8.2	8.0	9.1	8.6	9.1	9.7	8.5	
Clear-Sky Solar Irradiance	taub	0.372	0.403	0.434	0.499	0.467	0.466	0.487	0.494	0.465	0.417	0.383	0.383		
	taud	2.184	2.105	2.071	1.941	2.105	2.166	2.120	2.088	2.117	2.201	2.223	2.154		
	Ebn at Noon	225	239	250	243	257	257	250	243	238	233	219	204		
	Edh at Noon	30	39	45	56	49	46	48	48	43	35	29	28		
All-Sky Solar Radiation	RadAvg	354	590	997	1509	1885	2095	2039	1835	1371	822	442	298		
	RadStd	61	102	102	158	178	113	122	91	118	81	57	39		